LexAnalyzer · End-to-End System Flow & Architecture

Complete Lifecycle: Ingestion → OCR → Dense Vector Indexing → AI Auditing → Interactive Q&A;

1. Executive Overview

LexAnalyzer is an enterprise legal intelligence platform engineered for lawyers, legal auditors, and contract managers. It eliminates manual contract reviews by combining **optical character recognition (OCR)**, **retrieval-augmented generation (RAG)**, and **multi-tenant local vector stores** to uncover hidden risks, missing clauses, and ambiguous terms in seconds.

2. End-to-End Project Workflow (Start to Finish)

Every legal contract processed by LexAnalyzer flows through six deterministic stages:

• Step 1: Secure Multi-Tenant Authentication

Users register and log in via JWT authentication (RFC-7519) and bcrypt salt hashing. Every document, index, and audit query is strictly scoped to the authenticated user ID. Complete data privacy is enforced across tenants.

• Step 2: Multi-Format Ingestion & Tesseract OCR

Users upload native PDFs or scanned images (PNG, JPG, TIFF, WEBP, BMP). If an image or scanned document is detected, the automated OCR pipeline executes Tesseract OCR to extract high-accuracy text and wraps it into a standardized PDF.

• Step 3: Document Chunking & Dense Vectorization

PyMuPDF segments the document into semantic text chunks while preserving page numbers. Local HuggingFace Sentence-Transformers (*all-MiniLM-L6-v2*) generate 384-dimensional dense embeddings without transmitting confidential data to external vector APIs.

• Step 4: Vector Index Creation (FAISS)

Facebook AI Similarity Search (FAISS) builds a local on-disk index per document. This enables sub-millisecond nearest-neighbor search for precise clause retrieval.

• Step 5: Multi-Engine Legal Auditing

- **Contract Blind-Spot Detector:** Scans for omissions, ambiguous wording, missing exhibits, and undefined terms.
- **Clause Risk Analyzer:** Classifies indemnification, liability, and termination clauses into Low, Medium, and High risk.
- **Executive Summarizer:** Generates concise plain-English summaries and key business takeaways.

• Step 6: Interactive Q&A; & Audit Export

Legal counsel asks natural language questions in Document Chat. The system retrieves exact context chunks with page citations. Findings and gap analyses can be exported to Markdown (.md) or printed as formal audit reports.

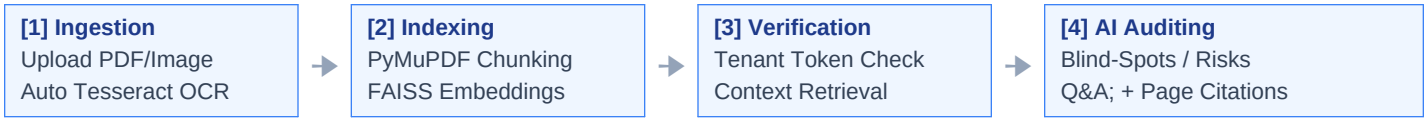
3. System Architecture & Component Design

LexAnalyzer is built on a modern decoupled architecture separating presentation, orchestration, AI retrieval, and isolated tenant storage.

Layer	Component / Tech	Function & Implementation Details
Frontend (Client)	React 18 · Vite Vanilla CSS · Lucide	• Responsive single-page application with custom dark navy glassmorphic design. • Zero bulky CSS framework overhead; dynamic filtering, instant live search, and audit badges.
API Gateway & Security	FastAPI · Uvicorn JWT · Bcrypt	• Asynchronous REST endpoints with strict Pydantic v2 validation. • Bearer JWT authorization middleware enforcing multi-tenant isolation on every document and query.
OCR & Parsing	PyMuPDF (fitz) Tesseract OCR	• Dual-mode extractor: parses digital text streams with page bounding or invokes local Tesseract OCR on scanned contract images.
Embeddings & Search	all-MiniLM-L6-v2 FAISS Vector DB	• Local 384-dimensional dense semantic vectors. Fast L2/Inner Product search per tenant index without external third-party vector databases.
AI Intelligence	LangChain · Groq OpenAI / Gemini / Heuristics	• High-speed inference using LLaMA-3.3-70B on Groq or OpenAI/Gemini. • Includes deterministic fallback heuristics ensuring 100% uptime even if API quotas expire.
Tenant Storage	JSON Store · Disk Tenant Directories	• Directory isolation: storage/documents/<user_id>/ and storage/index/<user_id>/. • Prevents cross-user data leakage at the operating system filesystem level.

4. Technical Data Flow Pipeline

Visual representation of data transformations from raw input to verified legal insights:



5. Visual Walkthrough & Interface Tour (Part 1)

Screenshots captured directly from the live LexAnalyzer environment showcasing core workflows:

Feature 1: Multi-Format Document Center & OCR Ingestion

- Drag-and-drop ingestion accepting digital PDFs and scanned images (PNG, JPG, TIFF, WEBP, BMP).
  - Auto-detects scanned documents, triggers OCR text extraction, and organizes indexes per tenant.

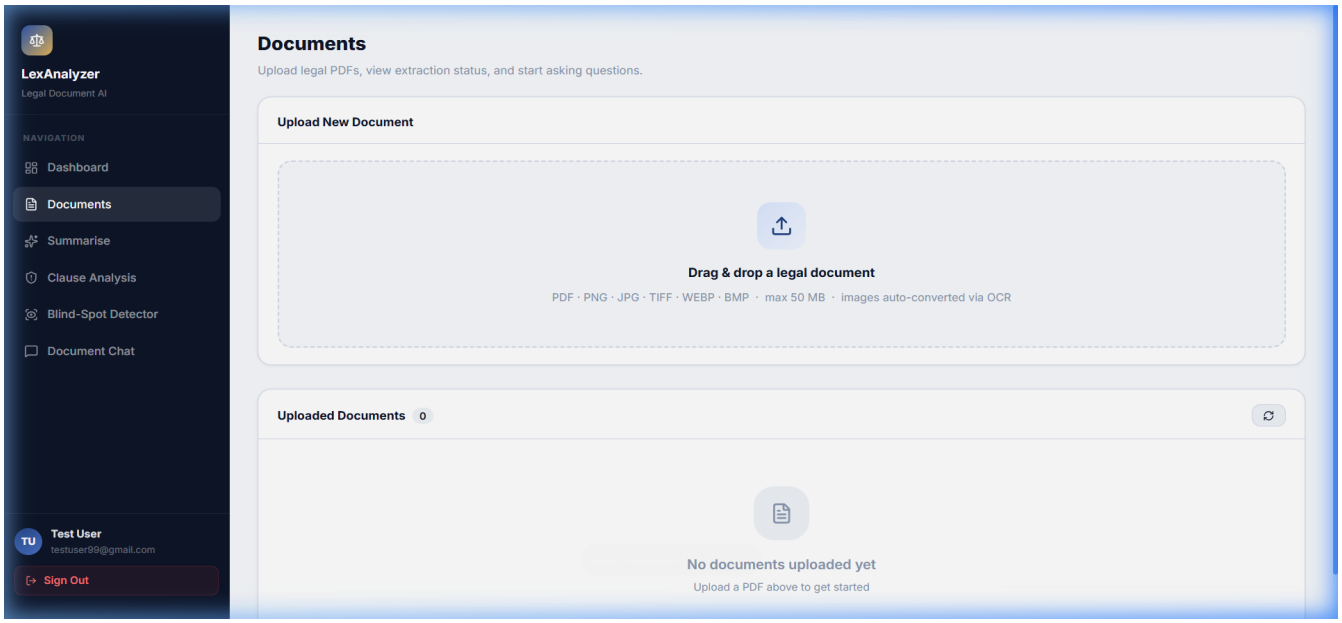


Figure 1: Document Upload & Ingestion Center with automatic OCR pipeline.

Feature 2: Interactive Document Chat with Verifiable Citations

- Natural language conversational interface grounded in indexed contract text.
  - Exact page-numbered citations prevent LLM hallucination and facilitate instant human review.

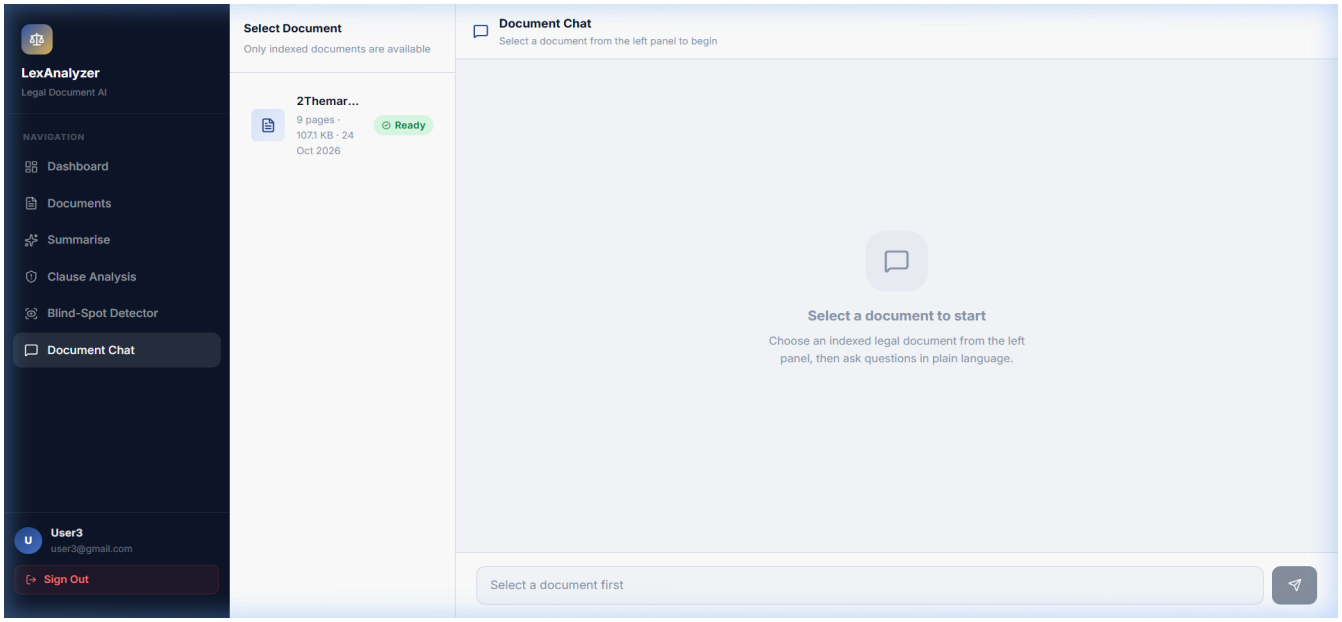


Figure 2: Natural Language Document Chat powered by RAG vector search.

6. Visual Walkthrough & Interface Tour (Part 2)

Feature 3: Contract Blind-Spot & Gap Detector

- Automatically uncovers what is *missing*: unattached exhibits, undefined terms, and vague deadlines.
  - Categorizes gaps by severity (High, Medium, Low) and suggests specific human review questions.

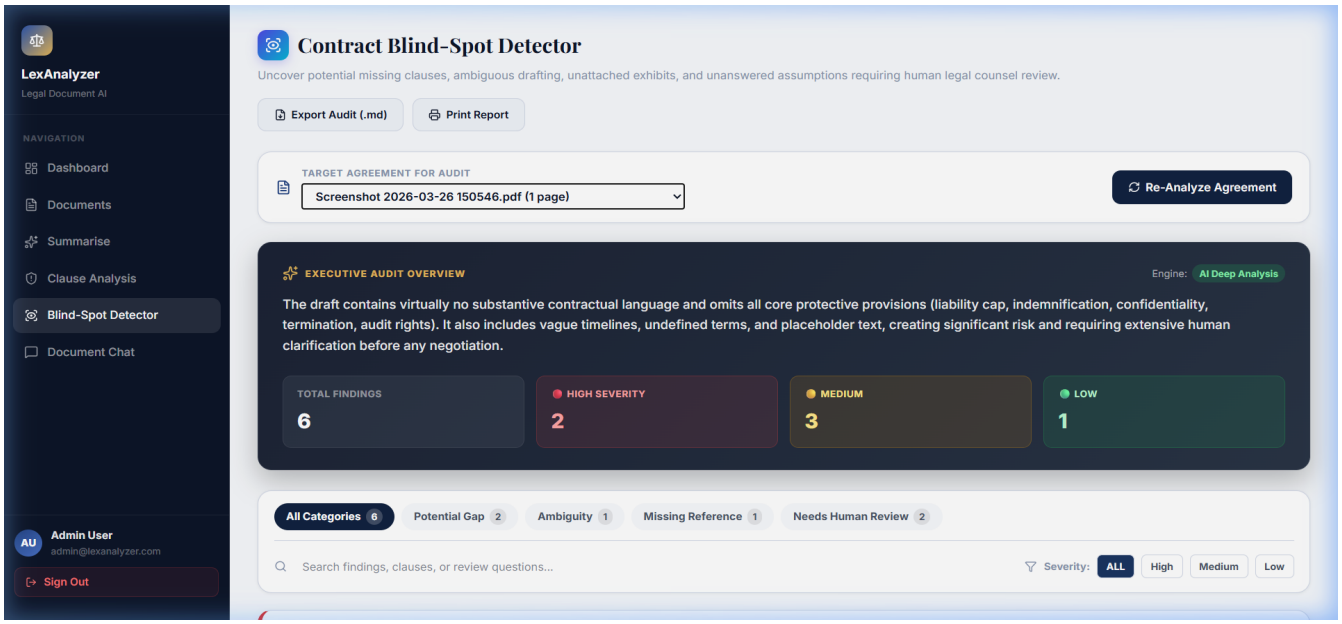


Figure 3: Contract Blind-Spot Detector identifying omissions, ambiguities, and missing exhibits.

Feature 4: Unified Analytics & Quick Action Dashboard

- Real-time overview of indexed agreements, audited clauses, and detected risk distributions.
  - Fast shortcuts to summarize documents, audit blind-spots, or query specific clauses.

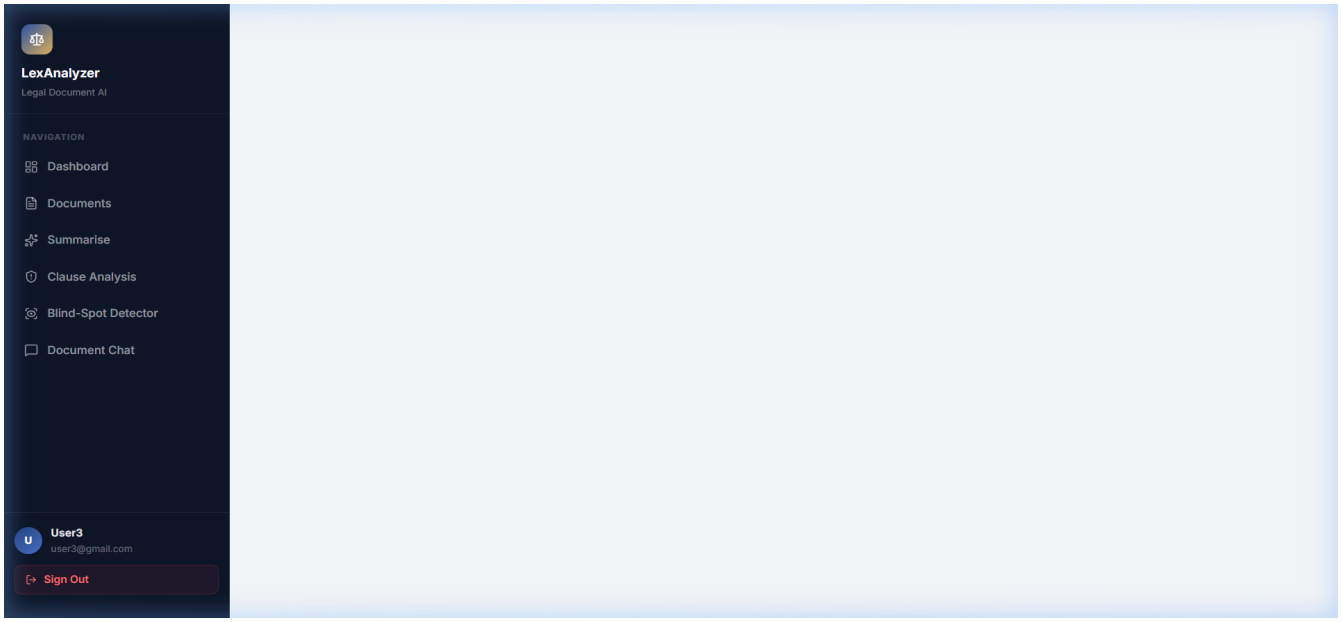


Figure 4: Executive Dashboard displaying document metrics, risk analytics, and quick links.